Fenugreek diseases

1. Root rot and web blight/ root rot - *Rhizoctonia solani*
2. Powdery mildew- *Erysiphe polygoni*, *Leveillula taurica*
3. Cercospora leaf Spot : *Cercospora traversiana*,
4. Downy mildew- *Peronospora trigonellae*

Fenugreek Cercospora leaf spot- *Cercospora traversiana*

Fenugreek Cercospora leaf spot- *Cercospora traversiana*

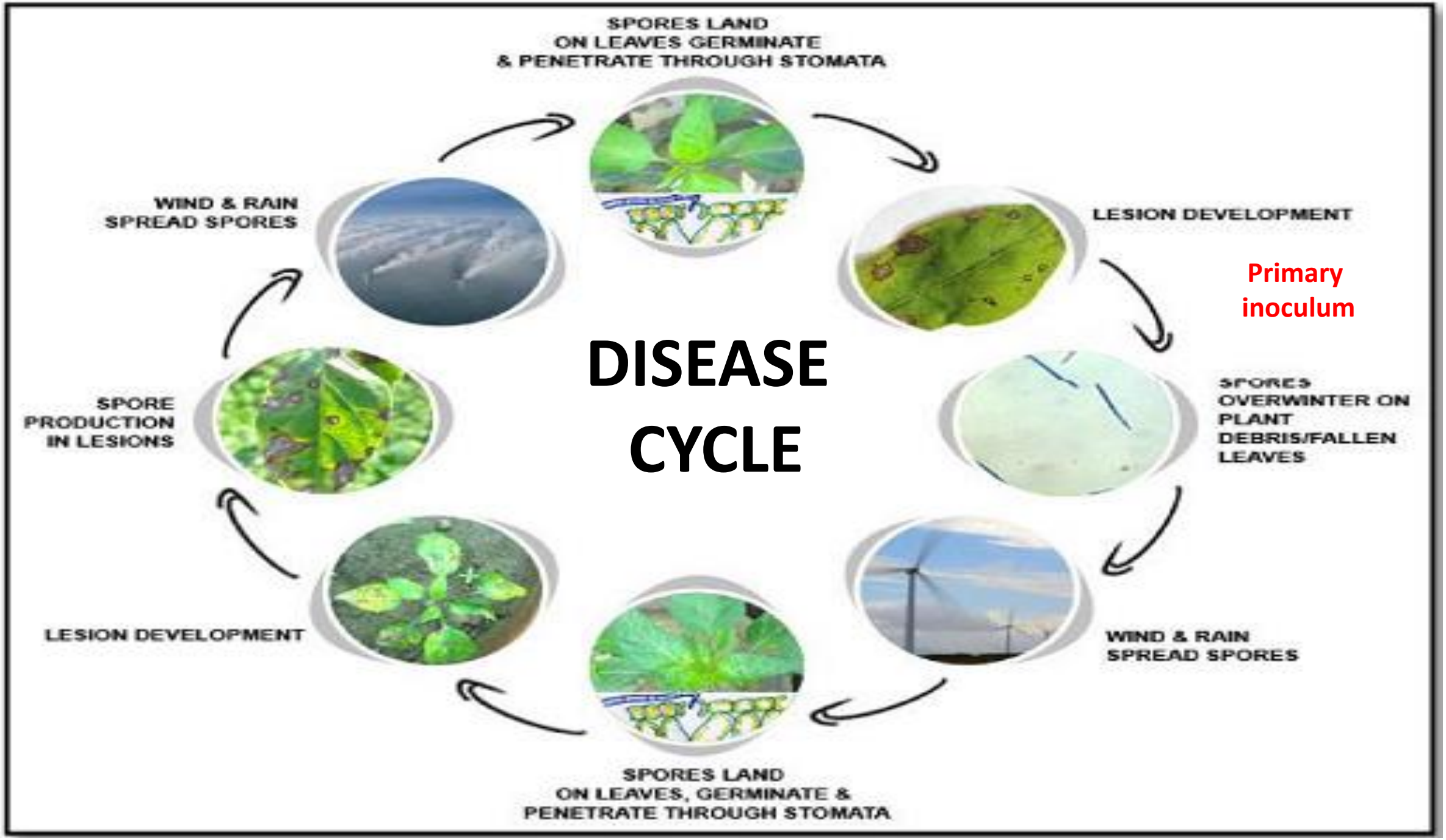
Symptoms

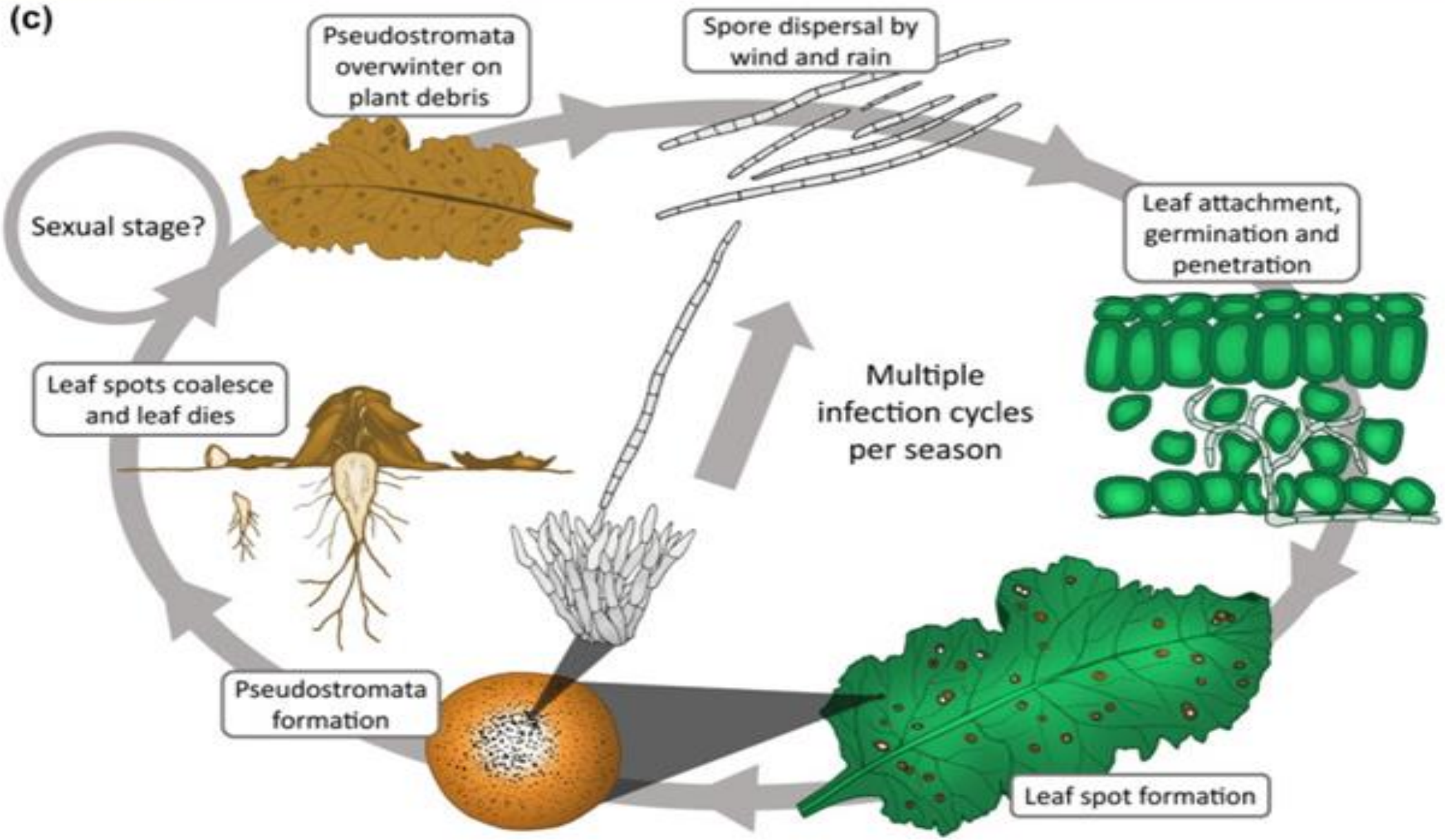
- The disease was uniformly distributed throughout the plot grown for seed
- **Leaves and stem :** First symptom appears as **large spots on leaves and stem**
- On individual plants only the upper few leaves remained, giving the appearance of tufts of green leaves scattered through out the plot.
- On older leaves lesion size had increased significantly and **sporulation was evident,** sometimes giving the lesion a whitish cast.
- Once the **young pods get infected the fungus infects the immature seed .**
- Seed: **Purple seed discolouration of seed coat** and deterioration in yield

Fenugreek Cercospora leaf spot- *Cercospora traversiana*

Symptoms contd…

- The necrotic areas were sharply defined and most lesions were surrounded by a **yellowish halo T**
- **Stem and pod infection also were severe.**
- The internally seed borne but **does not affect germination, but cause post emergent losses** and transmitted from seed to seedlings.
- **Discoloration of affected part of the pods,** if infection severe the pods get **twisted or shrunken, even the youngest leaves wilt and die.**


SPORES LAND
ON LEAVES GERMINATE
& PENETRATE THROUGH STOMATA
LESION DEVELOPMENT
Primary inoculum
SPORES OVERWINTER ON PLANT DEBRIS/FALLEN LEAVES
WIND & RAIN SPREAD SPORES
SPORES LAND
ON LEAVES, GERMINATE &
PENETRATE THROUGH STOMATA
LESION DEVELOPMENT
SPORE PRODUCTION IN LESIONS
WIND & RAIN SPREAD SPORES
DISEASE CYCLE


(c)
Pseudostromata overwinter on plant debris
Spore dispersal by wind and rain
Leaf attachment, germination and penetration
Sexual stage?
Multiple infection cycles per season
Leaf spots coalesce and leaf dies
Pseudostromata formation
Leaf spot formation

Fenugreek Cercospora leaf spot- *Cercospora traversiana*

Management

- Use of systemic fungicides
- Seed treatment with Captan and Thiram or Bavistin @2-3 g per kg of seed acts as first line of defense
- Follow field sanitation

Fenugreek Powdery mildew- *Erysiphe polygoni, Leveillula taurica*

Fenugreek Powdery mildew- *Erysiphe polygoni, Leveillula taurica*

Symptoms:

- **On leaves, stems and other foliar parts**
- white floury patches with conidia
- In severe cases **blighted foliage**
- Reduction **in pod number/plant, seed number/pod**
- Cleistothecia – crop residues at the end of the season

Fenugreek Powdery mildew- *Erysiphe polygoni, Leveillula taurica*

Pathogen:

- The disease is caused by *Erysiphe polygoni*, DC.
- **Fungus *Oidiopsis taurica*** (IS: *Leveillula taurica*) **has also been reported to be associated with this disease**.
- The mycelium of *Erysiphe* is generally fine and persistent, rarely thick.
- The haustoria develop as outgrowths from lobed swellings.
- The haustorium penetrates the cell wall and forms a sac in the epidermal cell.
- Conidiophores arise vertically and each bears several spores in chain.
- In **dry temperate zone, perithecia appear as sharp black specks, scattered on the surface of the white mycelium.**
- During following season, they disintegrate and liberate ascospores which germinate and infect the new crop.

Fenugreek Powdery mildew- *Erysiphe polygoni, Leveillula taurica*

Disease cycle and epidemiology:

- In some areas the ascocarps are formed for perennation.
- In other areas the pathogen survives **as mycelium and conidia** on the host and on other annual and perennial hosts.
- The disease is more prevalent in **dry weather and moderate temperatures.**

Fenugreek Powdery mildew- *Erysiphe polygoni, Leveillula taurica*

Management :

- **Sow the crop by the end of the October.**
- **Rest. cvs.** GC-39, UM-32 free,GC-7, GC-20,UM-34, Rmt-305

  Tolerant cvs. Hissar Suvarna, Hissar Madhuri
- With the initiation of the disease spray the crop with wettable sulphur (0.2%), or hexaconazole (0.05%) or dinocap (0.06%) or difenoconazole (0.04%) and repeat at 10-14 days interval.
- Dust sulphur at 25 kg/ha or foliar spray with wettable sulphur 2 g/lit at the time of appearance of disease.

Fenugreek downy mildew - *Peronospora trigonellae*

Upper surface (Adaxial) lower surface (Abaxial)

Downy mildew symptoms on upper leaf surface of a

Fenugreek downy mildew - *Peronospora trigonellae*

Symptoms:

- On leaves - **typical yellow stripes -turn brown**
- **Leaves turns yellow**
- A white to **dark-gray mycelium** was visible on the lower surface of the leaves, associated with **chlorotic spots on the upper leaf surface.**
- Plant dies if disease attack is severe
- **More incidence at flowering and pod formation**
- Oospores/ conidia

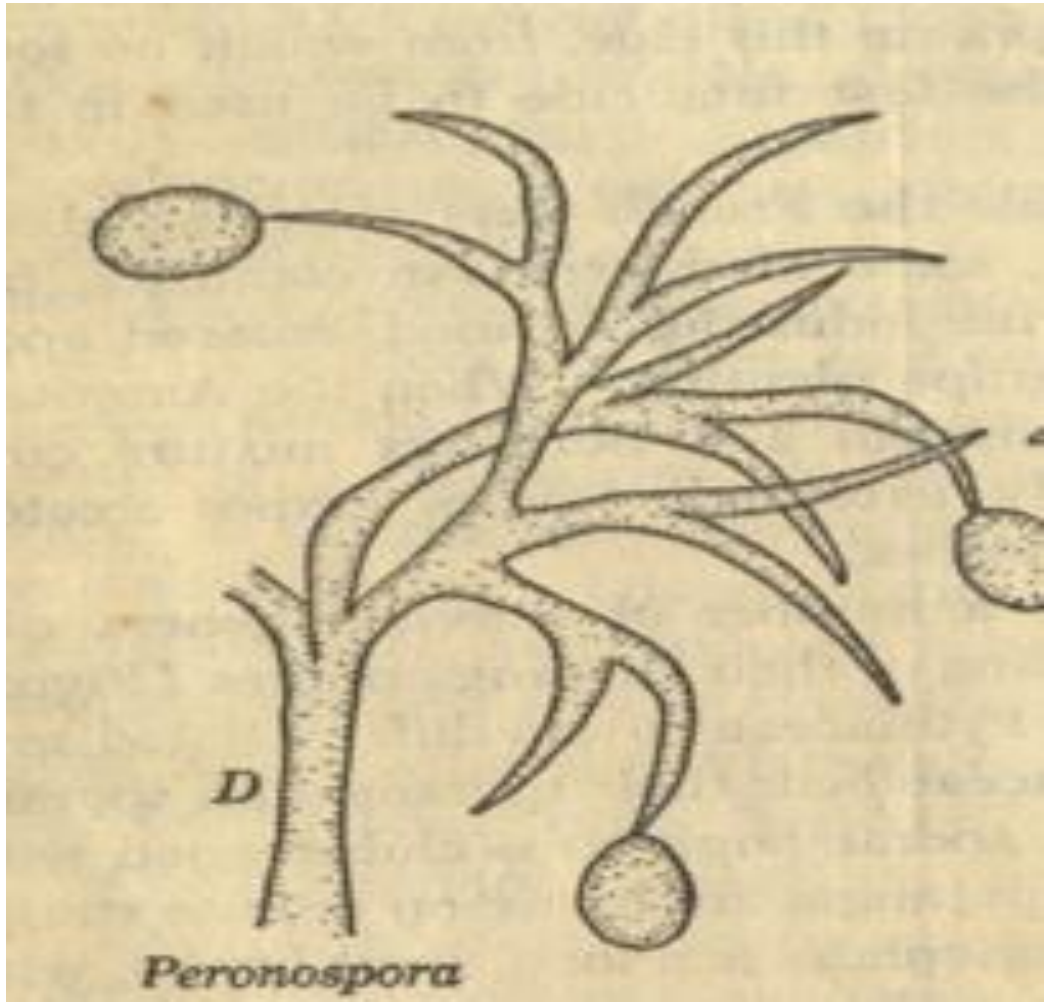
Peronospora spp.,

- Sporangiophores are dichotomously branched 2-7 times at acute angles, which finally end as sterigmata.

- Sterigmata are flexuous, gracefully curved and pointed with acute angles usually of equal length.

- Sporangia are oval and non-papillate sporangia are produced over the pointed sterigmata.

- Sporangia always germinate by germ tube, *i.e.*, sporangia behave as conidia.



Peronospora spp.,

Fenugreek downy mildew - *Peronospora trigonellae*

Management:

- Use adequate spacing when planting to avoid overcrowding
- Two/ three sprays with 0.2 percent solution of Dithane M45, Blitox @500-600 L / ha at intervals of 10-15 days reduces the infection
- Spray dithane Z -78 - 0.3% at 15 days interval

Root rot and web blight/ root rot - *Rhizoctonia solani*

Symptoms: Appear in two phases.

Root rot

- The disease cause severe losses in nursery . The pathogen cause water soaked lesions at collar region
- Softening of epidermal tissue at collar region due to water soaked lesions cause damping off
- As the disease reaches in advance stage, the stem becomes constricted at the base and plant collapses .
- Initial infection begins from infested soil as sclerotia of the fungus can survive in soil for long time
- The mycelium grows inside the tissue in all directions initiating secondary infections producing sclerotia on diseased part .

Fenugreek root rot and web blight/ root rot - *Rhizoctonia solani*

Web blight:

- Small, circular, water soaked, **tan coloured spots appear on leaves**.
- Dark brown, more or less circular, **slightly zonated and definitely sunken spots appear on pods** which in moist weather covered with hyphae and sclerotia of the fungus.
- **Tan brown to reddish brown discoloration of infected seed.**

Root rot and web blight/ root rot - *Rhizoctonia solani*

Management

- Seed treatment with brassicol .
- Cultivation of resistant varieties the best control measure
- Seed dressing and soil drenching with ***Rhizobium meliloti, Trichoderma banatum, T. harzianum*** can control the infection effectively.
- The Gram positive bacterium ***Bacillus subtilis*** can also be used effectively as a biological control agent for *R. solani*
- Carbendazim could be used as a seed and dry soil mix fungicide and that captan also could be used to drench the soil and kill the fungus.

Basella diseases

Cercospora leaf spot- *Cercospora basellae-albae, C. bataticola*- detailed study

1. Leaf blight - *Alternaria alternata*
2. Leaf and stem anthracnose – *Colletotrichum capsici, (Glomerella cingulata)*
3. Powdery mildew - *Erysiphe cichoracearum*
4. Leaf rot - *Pythium aphanidermatum*
5. Wilt and rot - *Sclerotium rolfsii*.
6. Brown leaf spot - *Curvularia lunata*
7. Leaf spot -*Pseudocercospora basella*
8. Damping-off *Rhizoctonia solani*

Basella Cercospora leaf spot- symptoms by *Cercospora beticola*

Basella - Cercospora leaf spot- *Cercospora basellae-albae, C. beticola*

Symptoms by *Cercospora basellae-albae*

- On leaves of different ages throughout the plant, although more spots are found on older leaves.
- Light green spots, approximately 0.5 mm in diameter, initially appear on both surfaces of the leaf.
- These lesions gradually increase in size and turn brown, with yellow green border.
- Two different zones are observed: the centre **is pale brown to whitish grey and the border is brown to purple black**
- Although **the veins limit the lesions,** those which are close to each other coalesce and **form blighted areas on the leaf**
- Severely blighted leaves turn yellow leading to defoliation.

Basella - Cercospora leaf spot- *Cercospora basellae-albae, C. beticola*

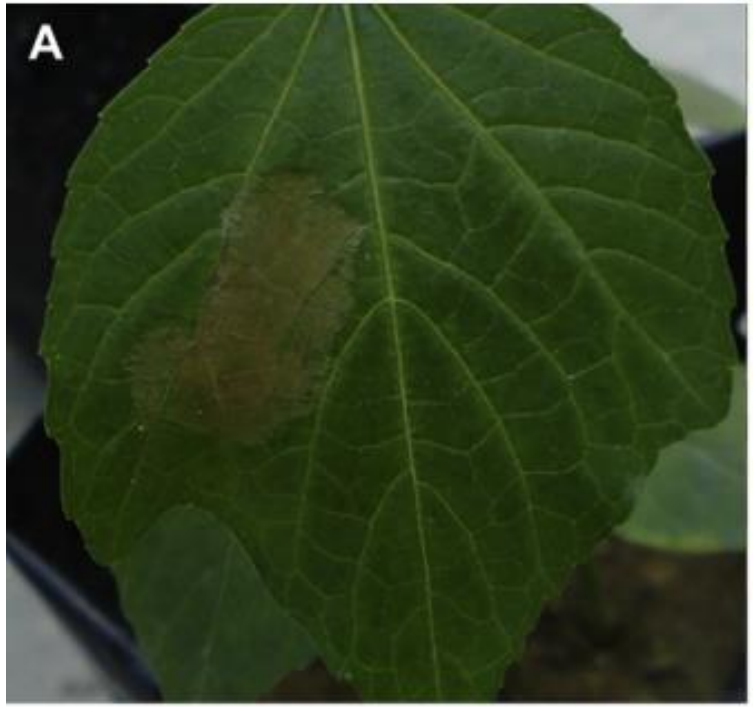
Symptoms by *C. beticola*

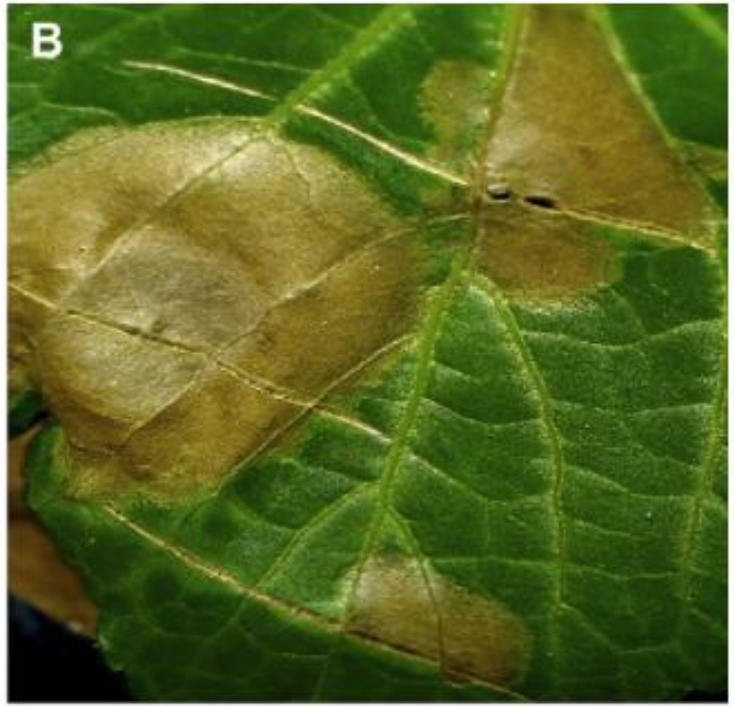
- With development of fungi and progress of disease, **white lesions surrounded by purplish brown border on the upper surface**.
- Corresponding **ventral surface of leaves showed pale brown lesion**.
- Sometimes several spots coalesce together to form large lesions

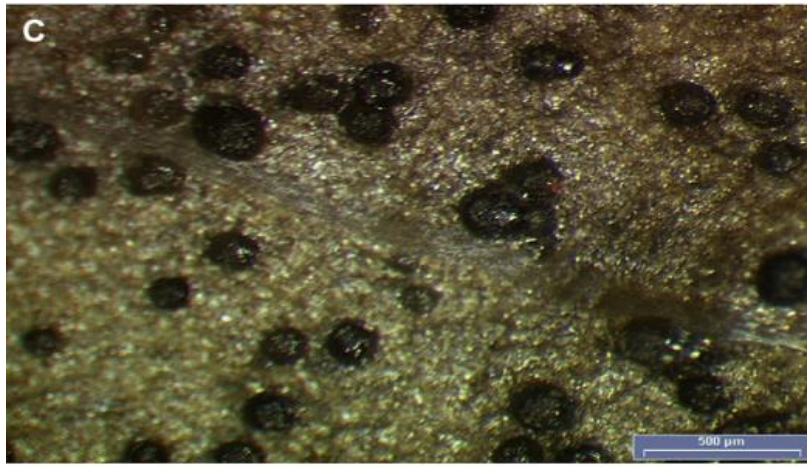
Hibiscus/Roselle diseases

1. **Phoma leaf spot- *Phoma exigua* (PS: *Leptosphaeria maculans*)**
2. Leaf spot -*Cercospora hibisci*
3. **Root and stem rot - *Phythophtora parasitica* var. *sabdariffae*.**
4. Root rot - *Botrytis cinerea*
5. Leaf fleck -*Phoma sabdariffae*
6. Root and seed rot -*Rhizoctonia solani*
7. Seed and stem rot -*Sclerotium rolfsii*
8. Powdery mildew -*Oidium abelmoschii*
9. Vascular wilt - *Fusarium oxysporum*
10. **Leaf Rust - *Kuehneola malvicola***
11. Hollyhock Rust - *Puccinia malvacearum*
12. Bacterial leaf spot-*Pseudomonas cichorii*

Hibiscus/Roselle Phoma leaf spot- *Phoma exigua*



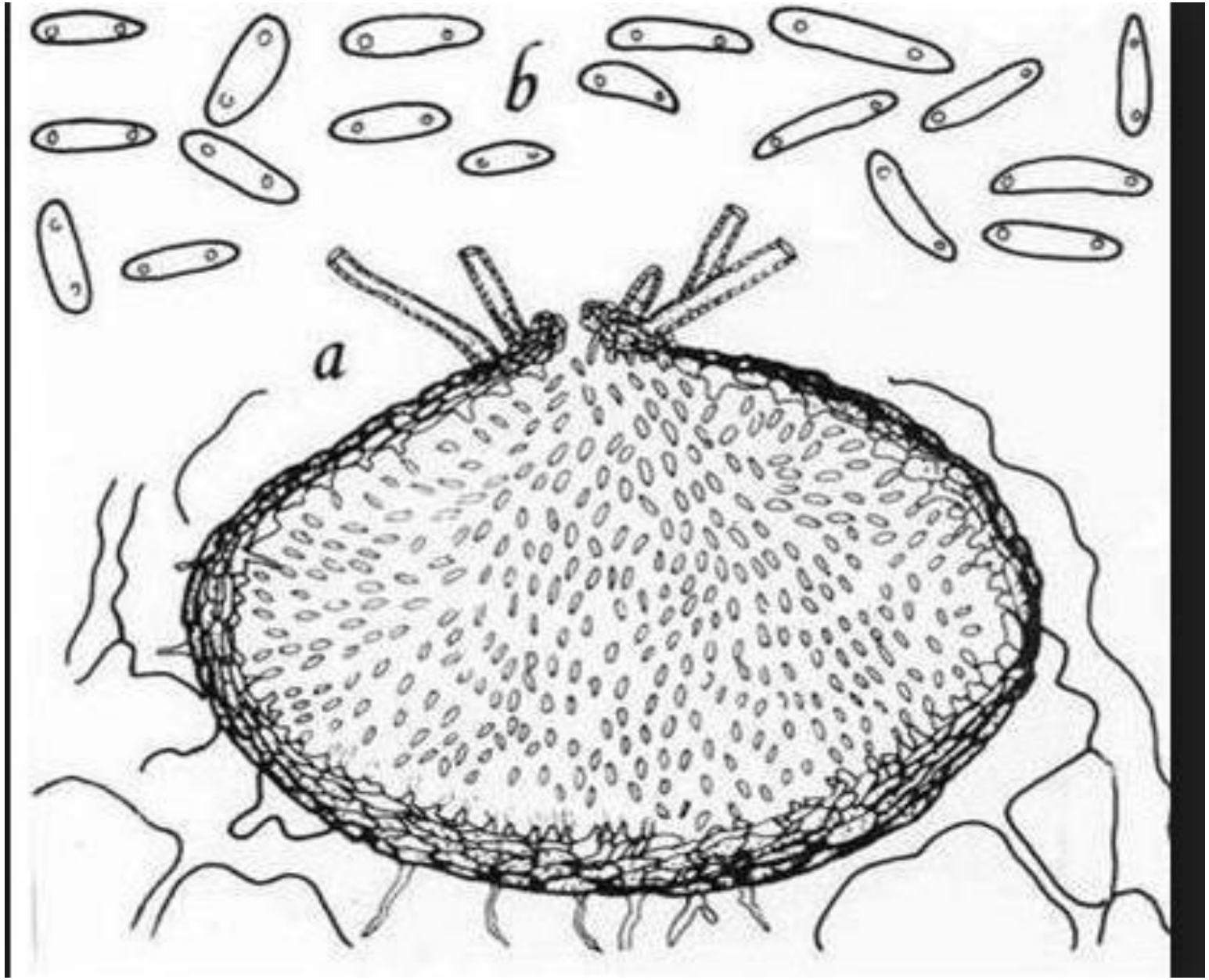




Hibiscus/Roselle Phoma leaf spot- *Phoma exigua*

Symptoms

- Symptoms started as **small circular spots which coalesced to form large, irregular lesions on the leaves**
- These **lesions later turned dark brown, with light brown margins.**
- Abundant black dots (pycnidia) appeared on the lesions, especially on the upper surface of the leaves in loose rings
- Severe infection **caused defoliation of the seedlings, resulting in a sparse crown.**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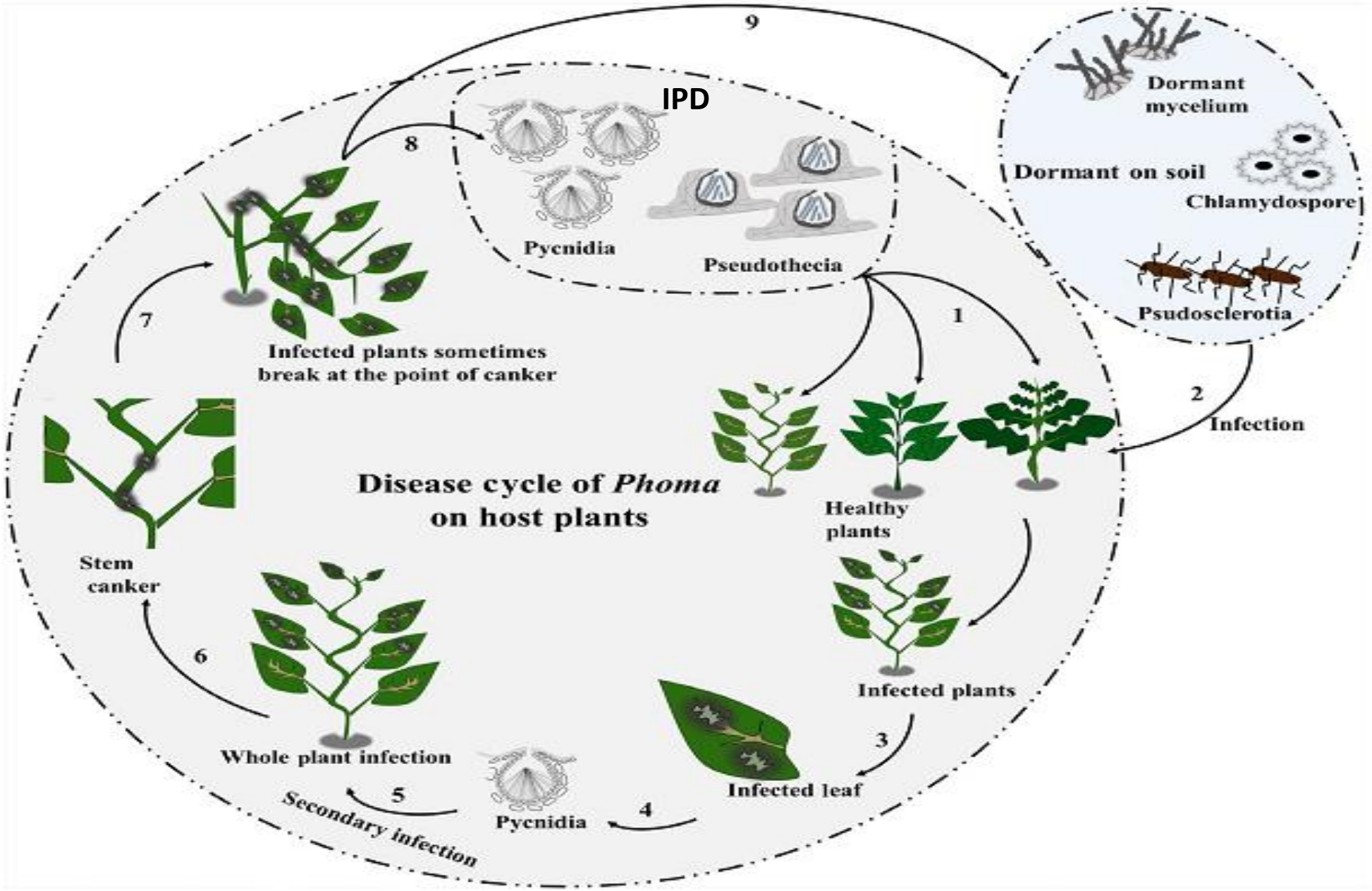

IPD
9
8
Dormant mycelium
Dormant on soil
Chlamydospore
Psudosclerotia
Pycnidia
Pseudothecia
1
2
Infection
Infected plants sometimes break at the point of canker
7
Disease cycle of *Phoma* on host plants
Healthy plants
Stem canker
Infected plants
6
3
Whole plant infection
Infected leaf
5
4
Pycnidia
Secondary infection

Hibiscus/Roselle Phoma leaf spot- *Phoma exigua*

Management:

- Removal and destruction of volunteer host plants and weeds
- Eradication of cruciferous weeds
- Crop rotation with non-host crops for 3 -4 years
- Soil Fumigation
- Soil solarisation
- Triazole fungicides are more effective for control of *L. maculans*

Hibiscus/Roselle root and stem rot- *Fusarium oxysporum, F. solani, F. equiseti, F. semitectum, Macrophomina phaseolina*

Hibiscus/Roselle root and stem rot- *Fusarium oxysporum, F. solani, F. equiseti, F. semitectum, Macrophomina phaseolina*

Symptoms

- Symptoms of naturally infected plants were characterized by cankers in the roots and the basal part of their stems adjacent to soil surface.
- The infected areas turn brown to black in color.
- Deterioration and destruction of secondary roots
- In severely infected cases, wilting and drying of plant

Hibiscus/Roselle root and stem rot- *Fusarium oxysporum, F. solani, F. equiseti, F. semitectum, Macrophomina phaseolina*

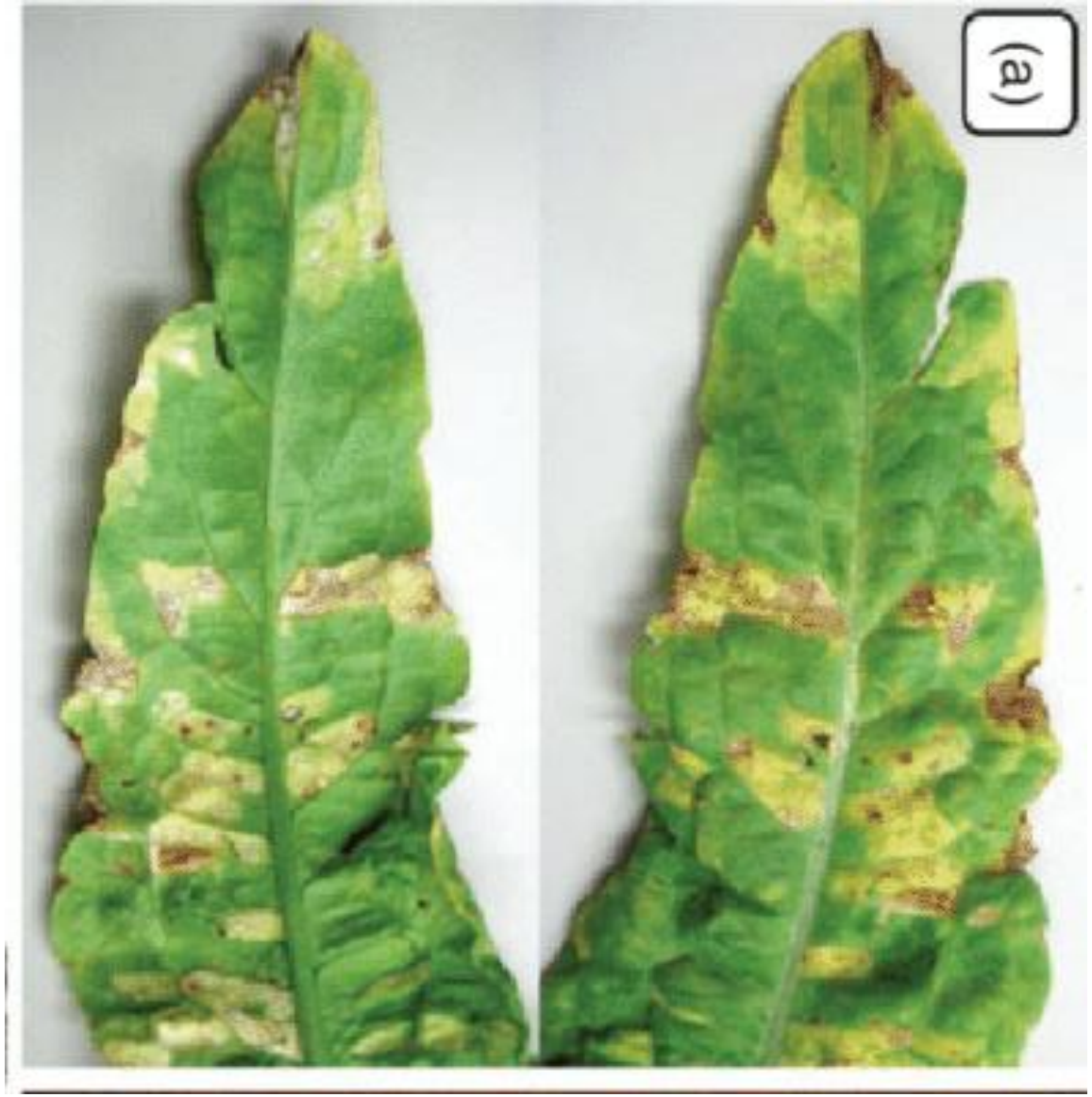
Symptoms contd…

- The leaves of the infected plants lose their turgidity, became flaccid, greenish yellow in color.
- The young tender shoots also wilt and die.
- In cross section of infected plant's stem and root, discolored brown areas appear as complete or incomplete ring representing the discolored vascular tissue.
- Later, the whole plant became brown with dried leaves.

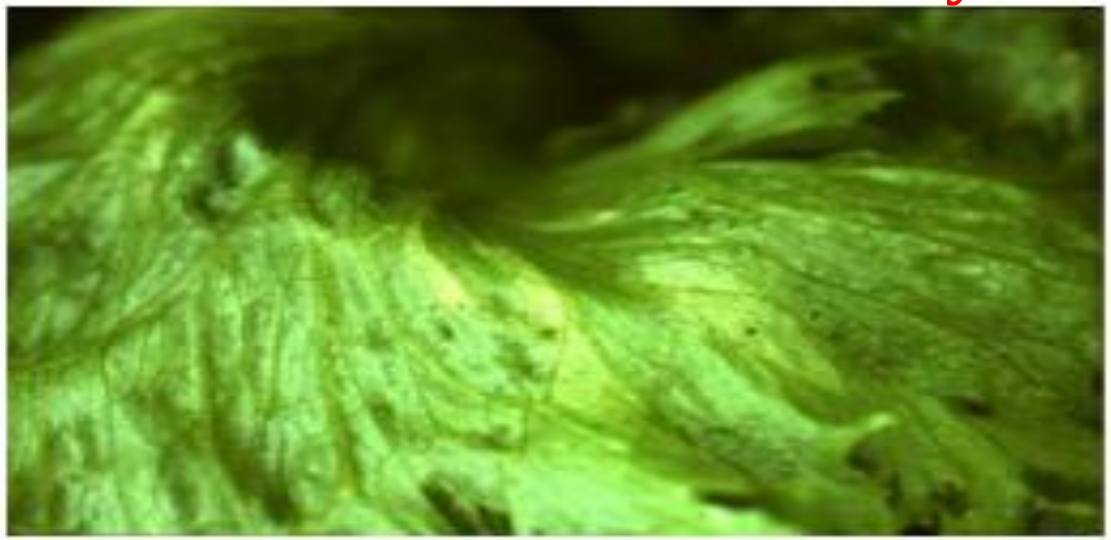
Lettuce diseases

1. Downy mildew - *Bremia lactucae*
2. Bacterial leaf spot - *Xanthomonas campestris* pv. *vitians*
3. Bacterial leaf spot and head rot - *Pectobacterium* (*Erwinia*) *caratovorum* sub sp *carotovarum*
4. Mosaic - *Lettuce mosaic virus* (LMV)

Lettuce downy mildew - *Bremia lactucae*



Lettuce downy mildew - *Bremia lactucae*



Downy Mildew on Lettuce (*Bremia lactucae*)

Figure 2. A white fluffy growth containing the spore producing structures and spores of *Bremia lactucae* typically develops from downy mildew infection sites on the underside of leaves.

Figure 1. Downy mildew infection sites confined by the veins of the lettuce leaf give these lesions an angular appearance. Light green to yellow spots on the upper leaf surface become necrotic and turn light tan to brown in color as they age. .

Lettuce downy mildew - *Bremia lactucae*

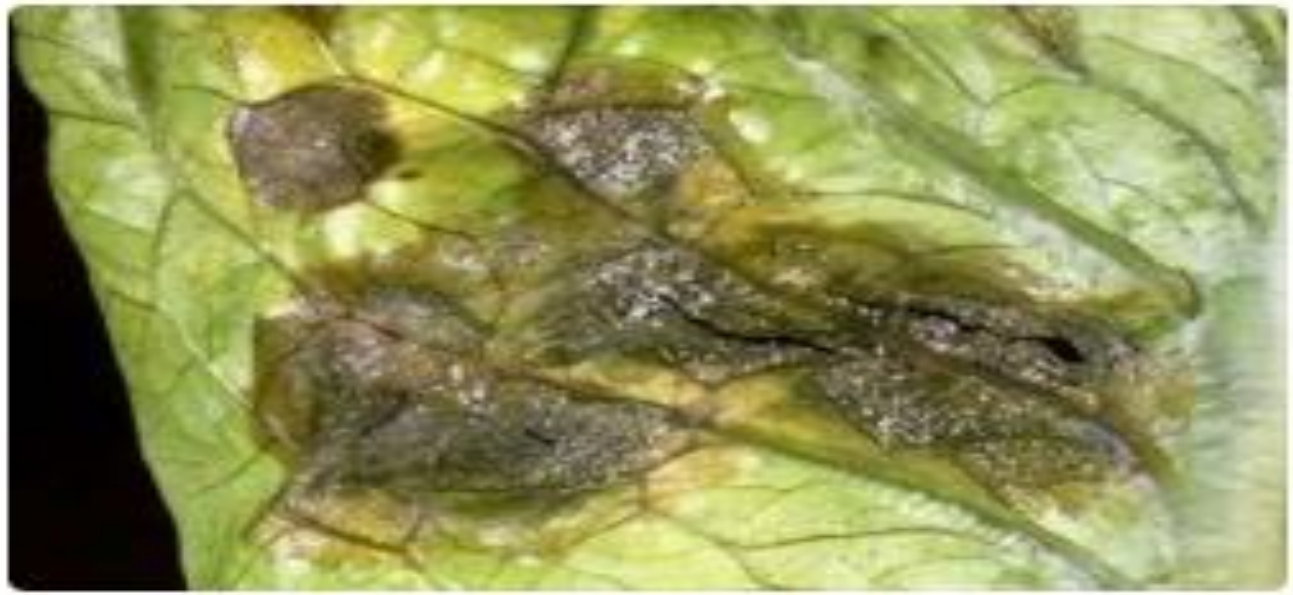


Photo 1. Brown patches on lettuce caused by downy mildew, *Bremia lactucae*.

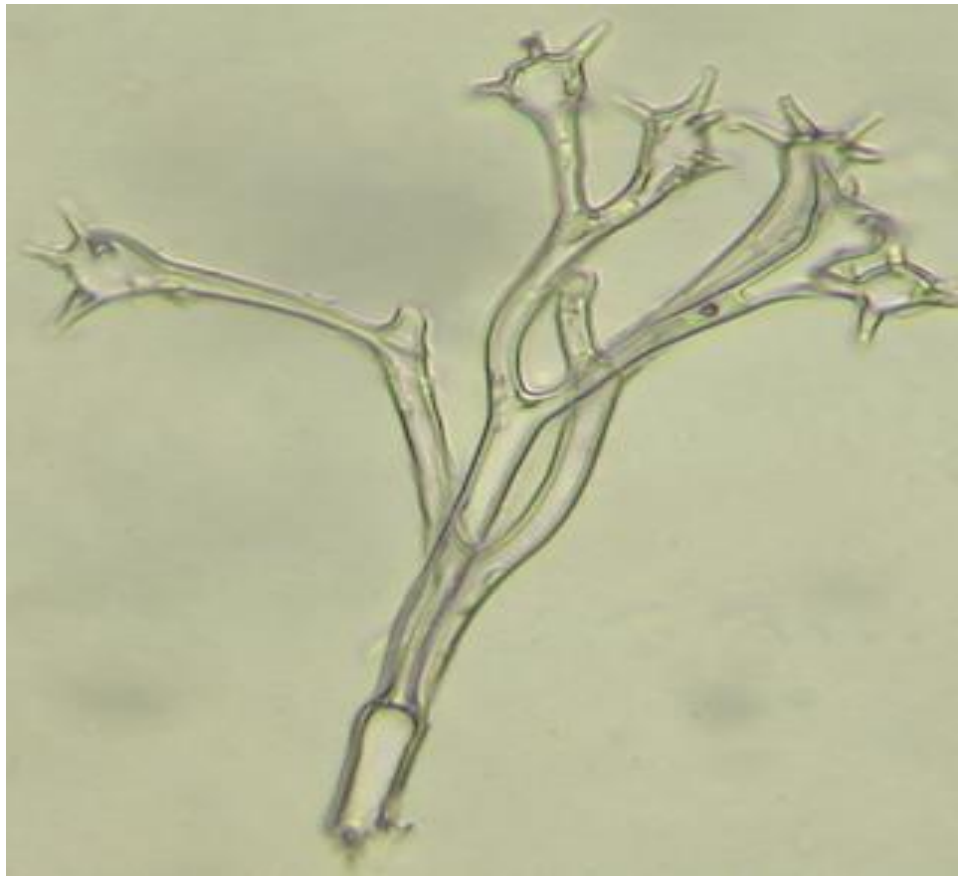
Lettuce downy mildew - *Bremia lactucae*

Symptoms:

- Leaf symptoms of downy mildew first show as angular, pale yellow patches which are delineated by leaf veins.
- The first leaves of the lettuce plant or older leaves close to ground usually are the first to show symptoms.
- The underside of the leaf opposite the yellow patch will show **white masses of spores** from 7 to 14 days after infection.
- As the spore mass grows larger, it takes on a white, **downy appearance**; hence the name downy mildew.
- As the disease spreads, the upper side of the leaves may also show the downy white fungus growth.
- Downy mildew can reduce yields and quality of the crop.

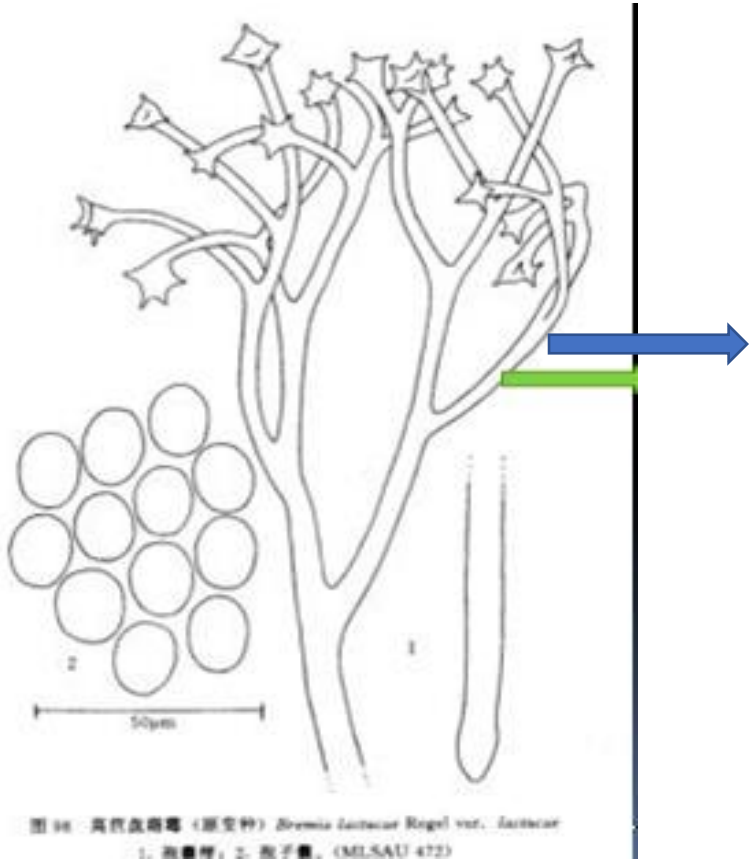
Bremia spp.,

- Sporangiophores are dichotomously branched.
- Tips of branches are expanded to cup shaped apophysis with four sterigmata bearing sporangia
- Sporangia are oval and papillate. Sporangia germinate by zoospores.
- Oogonial wall and oospore wall thin and unornamented



Bremia spp.,

Cup shaped structure.



Lettuce downy mildew - *Bremia lactucae*

Favourable conditions

- Downy mildew is favoured by **cool, wet and humid conditions**
- Favoured by **low temperatures 12-27^0 C**
- **High atmospheric humidity more than 90% following rain or dew**
- Night temperatures of 12°C-23°C will promote disease development, especially when accompanied by heavy dews, fog or precipitation.
- With night temperatures around 15°C and daytime temperatures around 25°C,

Lettuce downy mildew - *Bremia lactucae*

Management:

- **Remove infected leaves** - as the disease often only affects outer leaves the hearts can often be harvested and the rest of the plant removed and destroyed
- Seed treatment with Metalaxyl (2g/kg seeds
- Crop rotation to minimise the risk of infection from resting spores
- Do not overcrowd plants, because high humidity increases the risk of infection
- **Choose resistant cvs. Chartwell, Leny and Lettony.**
- **Prophylactic** - Mancozeb @ 0.25% before onset of the disease,
- **Curative -** Metalaxyl + Mancozeb @ 0.2%, and Dimethomorph @ 0.1% + Mancozeb or fosetyl-Al @ 0.2% or copper compounds at weekly interval at onset of the disease

Lettuce bacterial leaf spot - *Xanthomonas campestris* pv. *vitians*

Lettuce bacterial leaf spot - *Xanthomonas campestris* pv. *vitians*

Lettuce bacterial leaf spot - *Xanthomonas campestris* pv. *vitians*

Symptoms:

- An early symptom of bacterial leaf spot **is small** (less than 0.25 inch in diameter), **water-soaked leaf spots** on the older leaves of the plant.
- These lesions are typically **bordered by leaf veins and angular** in shape.
- Lesions quickly turn **black** (a diagnostic characteristic of this disease).

Lettuce bacterial leaf spot - *Xanthomonas campestris* pv. *vitians*

Symptoms contd….

- If the disease is severe, numerous lesions may coalesce, resulting in the **collapse of the leaf**.
- Older lesions dry up and become **papery** in texture, but retain the black color.
- Lesions rarely develop on newly developing leaves.
- Head **may rot** in severe infection

Lettuce bacterial leaf spot - *Xanthomonas campestris* pv. *vitians*

Management:

- Use healthy seed.
- Rotate with resistant crops
- Control alternate hosts in the crop and adjacent areas.
- Watering in the morning to reduce the time that leaves are wet and avoid overhead irrigation.
- Ensure good drainage.
- Avoid physical damage to Lettuce.

Lettuce Bacterial leaf spot and head rot - *Pectobacterium caratovorum* subsp *carotovarum*

Lettuce Bacterial leaf spot and head rot - *Pectobacterium caratovorum* subsp *carotovarum*

Slimy brown rot on the "head" of lettuce. Often these rots contain several species of bacteria causing soft rots, in addition to Pectobacterium carotovorum subsp. carotovorum.

Lettuce Bacterial leaf spot and head rot - *Pectobacterium caratovorum* subsp *carotovarum*

Symptoms:

- Bacterial soft rots affect the leaves and stems. **Water-soaked spots** occur on the outer leaves and form large brown **slimy areas** which progress to the inner leaves of the "head"
- The rots in the outer leaves cause the leaves **to wilt, and eventually the stem**, which results in collapse of the plant.
- Soft rot is important in the field during warm wet weather, and also important in transit and storage

Lettuce Bacterial leaf spot and head rot - *Pectobacterium caratovorum* subsp *carotovarum*

Management:

Before planting

Avoid planting in waterlogged soils and plant on raised beds.

During growth

- Do not plant lettuce too close to each other
- Mulch plants with, e.g., straw or dried grass
- Avoid over-head irrigation, use drip irrigation in preference.
- Avoid injury at the time of weeding
- Clean and disinfect equipment used in harvesting.
- Remove plants with signs of soft rot as soon as it appears, and burn or bury the plants deeply.
- Avoid harvesting lettuce when they are wet.

Lettuce Bacterial leaf spot and head rot - *Pectobacterium caratovorum* subsp *carotovarum*

Management contd….

After harvest:

- Handle plants carefully to avoid wounding the leaves.
- If possible, keep lettuce cool after harvest; **4ºC** is ideal.
- Collect crop debris and burn or bury it deeply in the soil before a new crop is planted.
- Crop rotation for 2-3 years with beans, cucumbers or tomatoes.

Lettuce mosaic - *Lettuce mosaic virus*

Lettuce mosaic - *Lettuce mosaic virus*

Lettuce mosaic - *Lettuce mosaic virus*

Symptoms

- Symptoms of lettuce mosaic vary greatly.
- Leaves of plants that are infected at a young stage are **stunted, deformed**, and (in some varieties) show a **mosaic or mottling pattern**.
- Such plants rarely grow to full size; head lettuce varieties that are infected early fail to form heads.
- Plants that are infected later in the growth cycle will show a different set of symptoms.

Lettuce mosaic - *Lettuce mosaic virus*

Symptoms contd….

- These plants may reach full size, but the **older outer leaves will be yellow, twisted, and otherwise deformed**.
- On head lettuce, the wrapper leaves often will **curve back away from the head.**
- Developing heads may be deformed. In some cases, **brown, necrotic flecks occur on the wrapper leaves**

Lettuce mosaic - *Lettuce mosaic virus*

Management

- A lettuce mosaic control program is a good example of an integrated way of controlling a plant disease.
- Plant lettuce seed that has been tested for the virus and that contains no infected seed per 30,000 seed tested.
- Only tested, clean seed can be planted.
- Remove potential virus reservoirs (pea, spinach) by practicing good weed control within and outside lettuce fields, and by ploughing down harvested lettuce fields in a timely manner.

Moringa diseases

Twig canker - *Fusarium pallidoroseum*

Moringa twig canker - *Fusarium pallidoroseum*

Symptoms

- Cankers observed in late spring to summer were usually small (1 to 3 cm long) and mainly occurred on the new growth strongly incited by intensive pruning.

- Pale orange **sporodochia** were evident on lesions

Moringa twig canker - *Fusarium pallidoroseum*

Management

- The disease can be controlled by pruning and burning all cankered and dead branches.
- Remove cankers from trunks and large branches during dry weather.
  - Treat the wound and all pruning cut with **disinfectants**
  - Wound dressing **@Bordeaux mixture**
  - Spray with **Carbendazim@ 1g/lit**